Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Arrival · Independent

1. Enzyme and substrate

Identify the enzyme and substrate in amylase acting on starch.

Enzyme = catalyst. Substrate = molecule the enzyme acts on.

2. Read the colour

At 60 s iodine is blue-black; at 90 s it stays orange-brown. Which sampled time first meets the endpoint?

The agreed endpoint is the first sampled time when starch is not detected.

3. Change and measure

State the independent and dependent variables, then name one condition to keep constant.

Change buffer pH; measure endpoint time. Controls include temperature and starting solution amounts.

4. Prepare the evidence

Why prepare the iodine wells and time headings before mixing?

Once amylase and starch are mixed, the reaction continues while equipment is being prepared.

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Exit Ticket · Independent

Evidence

These are example results at pH 7: 30 s blue-black; 60 s blue-black; 90 s orange-brown. Blue-black means starch was detected. Orange-brown means no starch was detected in that sample. The planned interval is 30 s. The amounts at the start match. The temperature and the endpoint rule match too.

1. State the first observed endpoint and the window in which it was reached.

Use “record ___ s; after ___ s and by ___ s”.

2. Give one way to protect a fair comparison across pH values and explain why it matters.

Use one matched condition: temperature, starting amounts or sampling interval.

Under the heading Exit ticket: answer the short questions in your book.